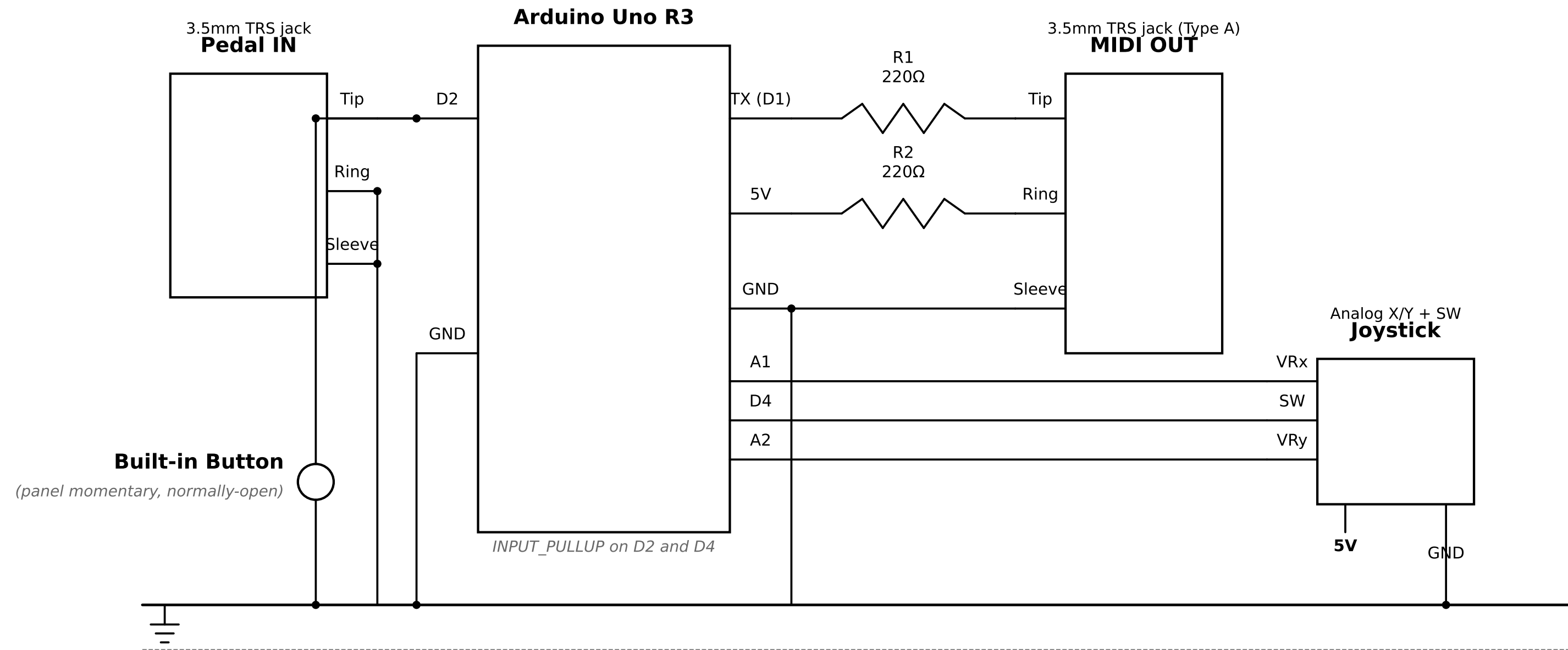


FM-1 Sustain Footswitch — Schematic

Arduino Uno R3 + TRS Pedal In + Built-in Button + TRS MIDI Out (Type A) + Joystick Pitch-Bend



Notes:

- Built-in Button is wired in parallel with the pedal jack (same D2/GND nodes) -- either one alone can trigger sustain, so an external pedal is optional
- Pedal ring + sleeve are tied together so a plain mono (TS) pedal plug still grounds correctly in the TRS jack
- D2 uses INPUT_PULLUP -- the button or pedal only needs to short D2 to GND when pressed (normally-open momentary)
- Type A TRS MIDI: tip = signal, ring = +5V current source, sleeve = ground
- R1 limits current through the FM-1's opto-isolated MIDI input; R2 sources the +5V loop current -- both 220Ω, matching the standard DIN MIDI-out circuit
- TX (D1) is shared with the USB-serial programmer -- disconnect the MIDI OUT jack (or at least the R1 lead) before uploading a new sketch
- If the FM-1 doesn't respond, it may expect Type B instead: swap Tip and Ring at the MIDI OUT jack (tip = GND, ring = signal, sleeve = GND)
- Joystick VRx (A1) drives pitch bend, VRy (A2) selects octave (down/center/up), SW (D4, INPUT_PULLUP) holds to sound a note and glide it -- release to stop
- The FM-1 ignores pitch bend sent to an already-sounding note, but applies it if sent right before Note On -- so the sketch rapidly retriggers the note (~15ms) with a fresh bend value each time to fake a smooth glide
- OLED display, mode switch, and audio-input pitch detector removed for now -- coming back later